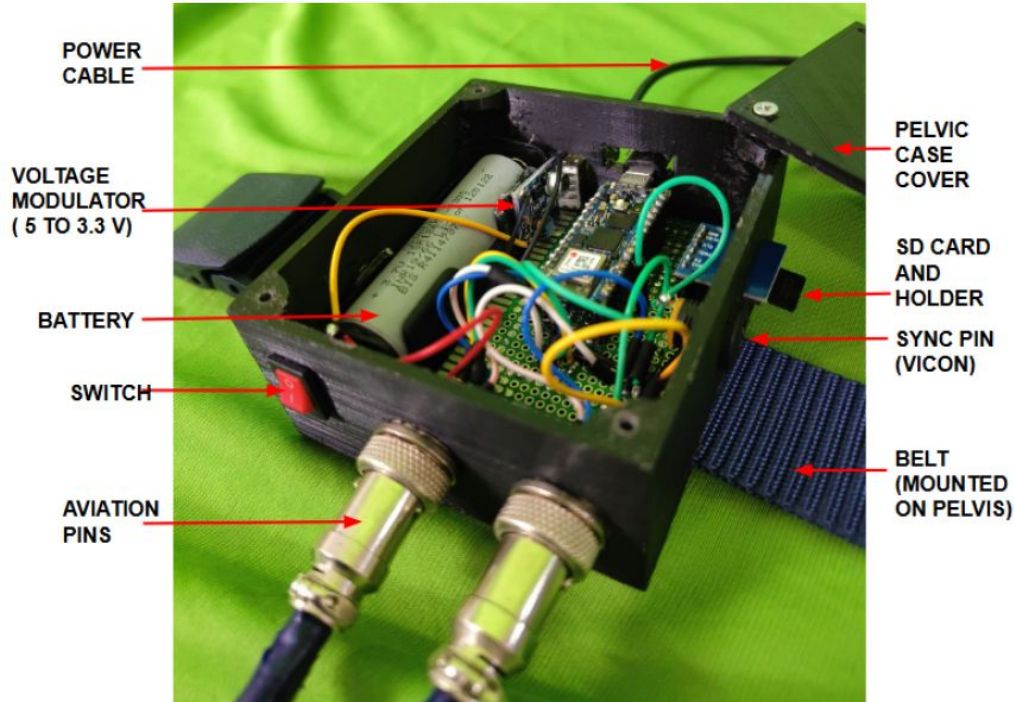


Hardware Prototype: Wearable IMU System (50 Hz)

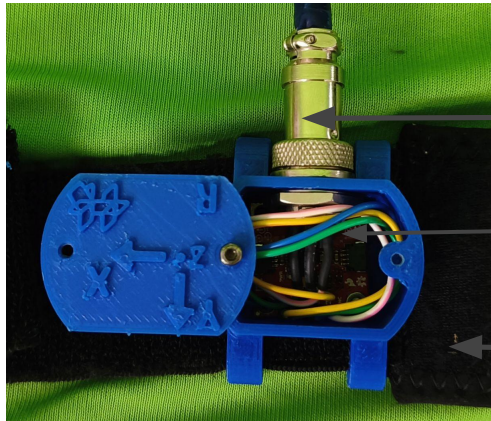
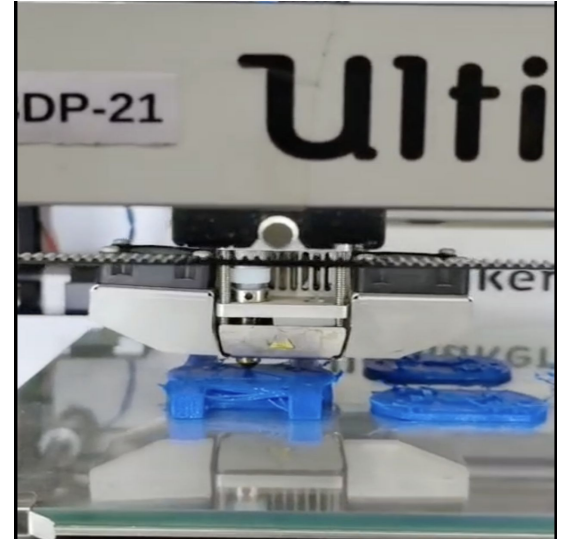
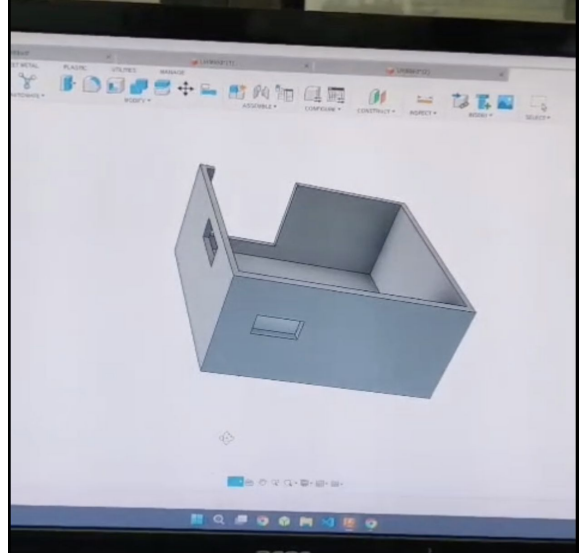
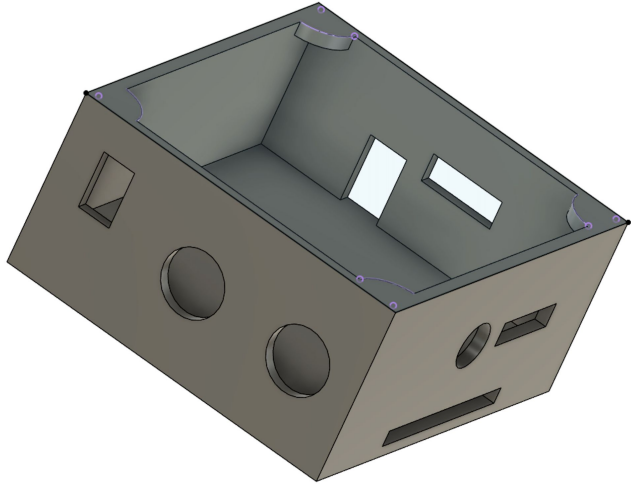
(a) Pelvic module: Arduino Nano **RP2040** microcontroller
| Sparkfun **ICM-20948** | 3.7 V Lithium-ion | I2C wiring (4
core) - SDA(data), SCL(clock), ground and voltage in.



(b) Full-body IMU, VR and Vicon marker setup



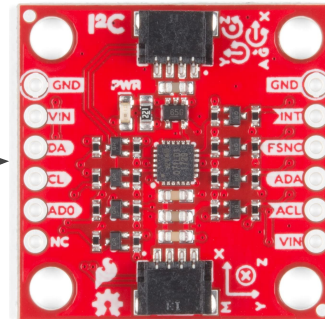
CAD DESIGN (Autodesk Fusion 360)



AVIATION PINS

IMU

BELT (MOUNTED ON SHANK)



9 DOF:

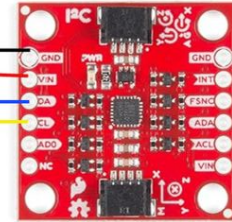
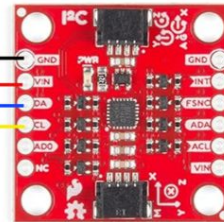
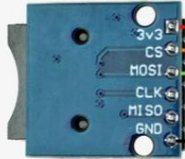
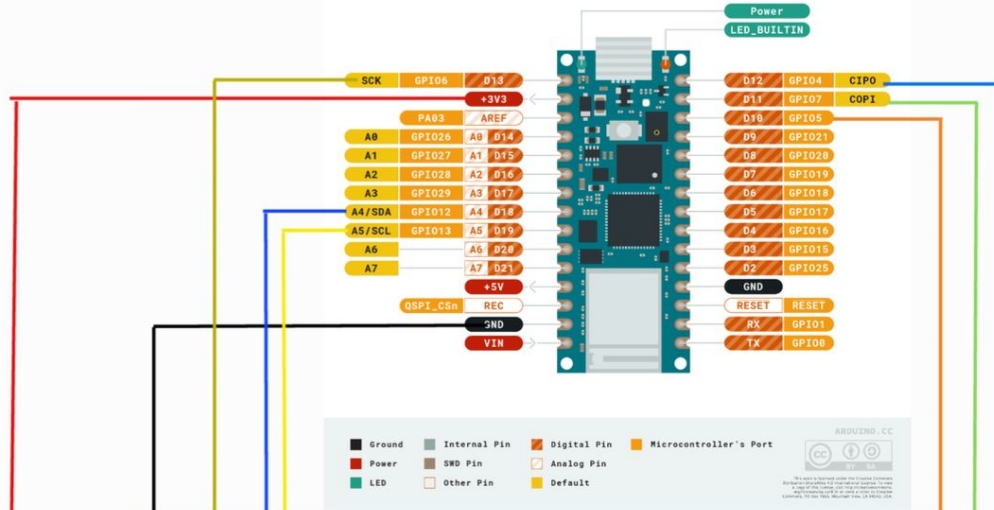
3 axis accelerometer

3 axis gyroscope

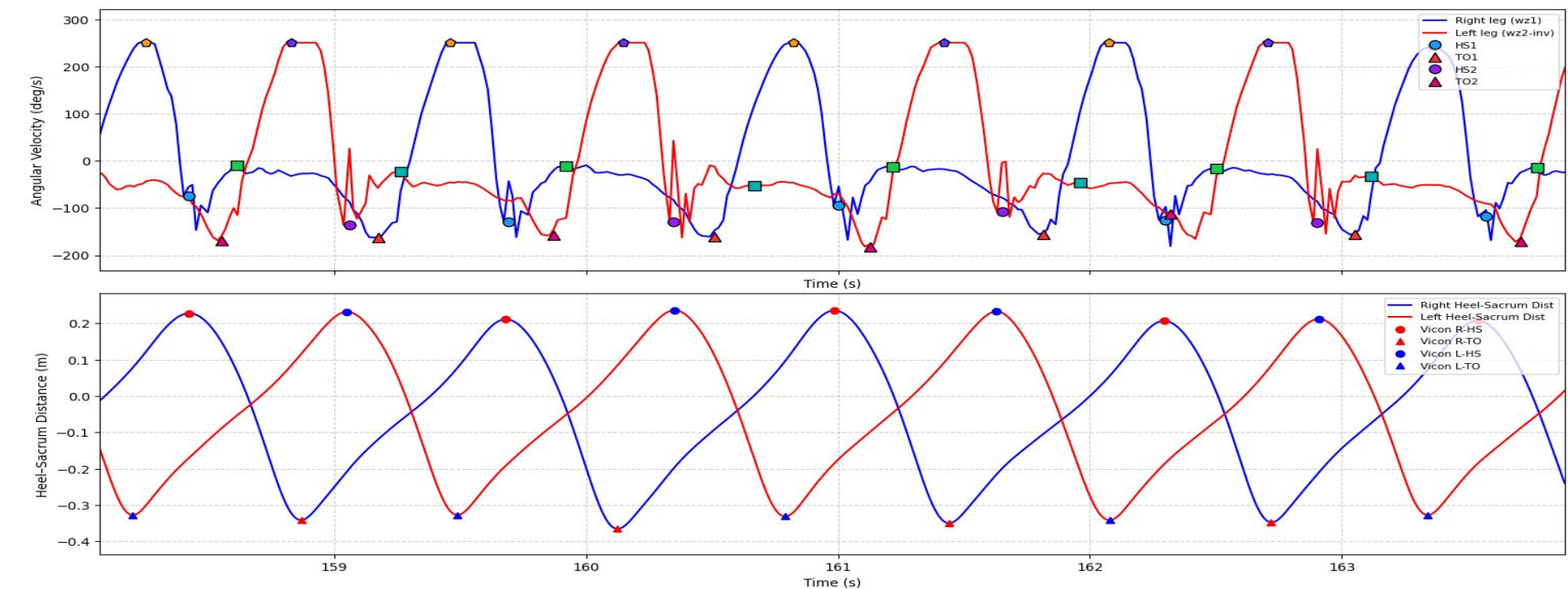
3 axis magnetometer



ARDUINO NANO RP2040 CONNECT

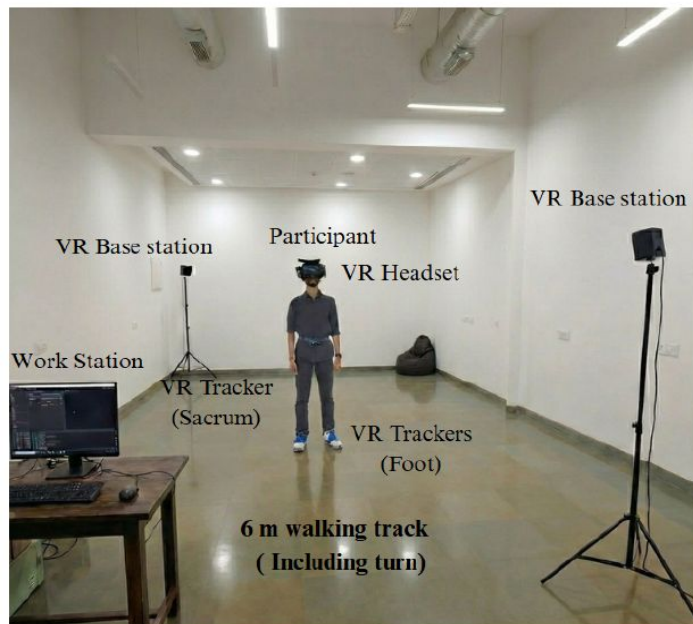


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE
1	time_ms	ind_HS1	ind_HS2	ind_LP1	ind_LP2	ind_MS1	ind_MS2	ind_TO1	ind_TO2	ind_ZC1	ind_ZC2	ax1	ay1	az1	ax2	ay2	az2	wx1	wy1	wz1	wx2	wy2	wz2	wz1	AxL	AyL	AzL	GxL	GyL	GzL	data_rate
2	23636	0	0	0	0	0	0	0	0	0	0	-104.98	-989.75	-191.41	10.99	-999.27	-65.67	-0.397	1.969	4.718	0.466	0.824	-0.313	0.313	0.980	-0.003	-0.180	-0.916	-0.183	-0.366	0.000
3	23655	0	0	0	0	0	0	0	0	0	0	-109.13	-983.89	-184.08	18.55	-999.27	-63.96	-0.405	1.885	5.046	0.221	0.672	-0.435	0.435	0.979	-0.008	-0.176	-0.366	-0.610	-0.549	0.000
4	23670	0	0	0	0	0	0	0	0	0	0	-109.13	-989.99	-193.60	5.37	-1008.79	-65.19	-0.382	1.496	4.656	0.481	0.359	-0.649	0.649	0.978	-0.009	-0.174	-0.183	-0.916	-0.671	0.000
5	23689	0	0	0	0	0	0	0	0	0	0	-114.99	-987.79	-190.43	16.85	-1002.93	-68.60	-0.084	1.405	4.954	0.733	-0.794	-1.267	1.267	0.974	-0.009	-0.172	0.366	-1.587	-1.038	0.000
6	23704	0	0	0	0	0	0	0	0	0	0	-115.23	-989.50	-197.27	17.09	-1002.20	-70.31	-0.237	1.511	4.534	0.771	-1.198	-0.878	0.878	0.973	-0.008	-0.170	1.038	-2.258	-1.282	0.000
7	23718	0	0	0	0	0	0	0	0	0	0	-111.08	-985.60	-189.70	25.63	-1006.35	-69.82	-0.450	1.496	4.718	0.595	-1.107	-0.420	0.420	0.975	-0.008	-0.167	1.038	-2.686	-1.221	0.000
8	23737	0	0	0	0	0	0	0	0	0	0	-122.80	-981.69	-186.77	22.22	-995.61	-65.92	-0.351	1.290	4.298	0.298	-0.427	-0.328	0.328	0.979	-0.007	-0.167	1.343	-3.296	-1.282	0.000

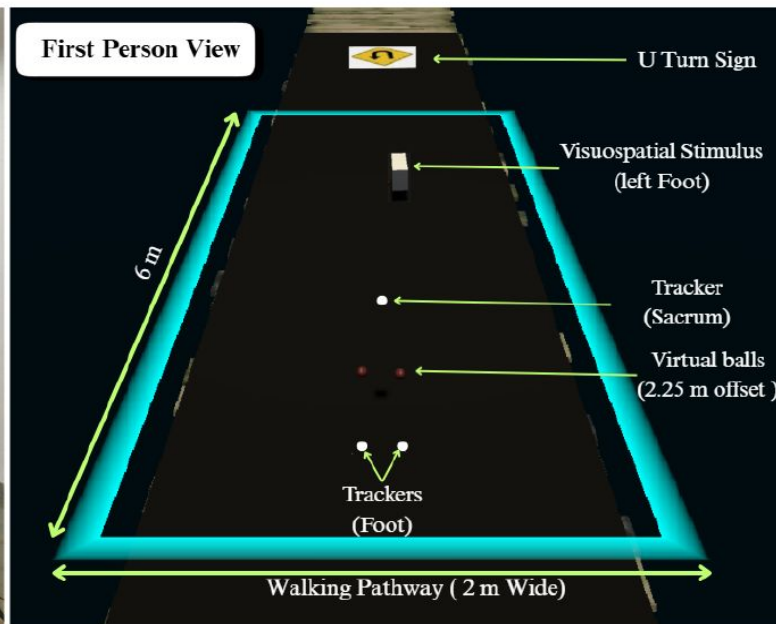


Hardware Prototype: VR Setup (90 Hz)

- Virtual bridge environment: to simulate a constrained walking pathway
- Walking trials: **Baseline(no perturbations)** and **perturbation trials** [White box: left, Green box: Right]
- Trackers: real-time positional data for reconstructing motion



(a) VR Sensor Setup



(b) Experimental Setup

VR Environment and Unity post processing codes credits: Chetan Sharma(HCR Lab)

Hardware Prototype: Vicon (Plug-in-Gait Model) (200 Hz)

- CALIBRATION
- 8 infrared cameras: track reflective markers
- High-precision 3D kinematic data: **gold standard**
- Accurate detection of gait events, allowing computation of spatiotemporal gait parameters.

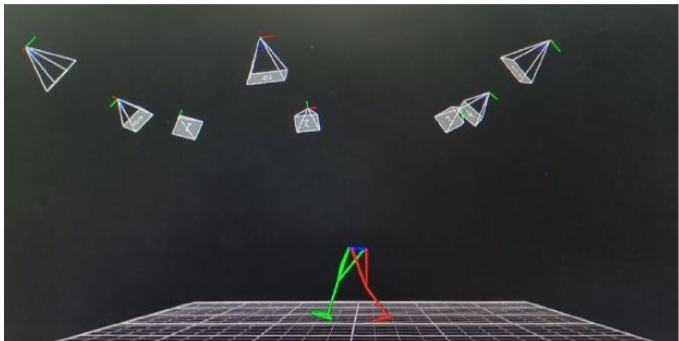
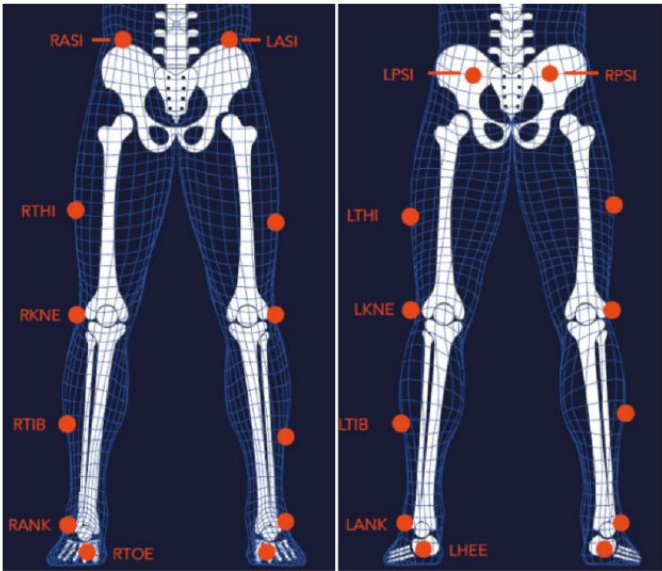


Table 2.3: Plug-in Gait lower-body marker configuration

Region	Right Marker	Left Marker	Anatomical Location
Pelvis	RASI	LASI	Anterior Superior Iliac Spine (ASIS)
Pelvis	RPSI	LPSI	Posterior Superior Iliac Spine (PSIS)
Thigh	RTHI	LTHI	Mid-thigh, lateral surface
Knee	RKNE	LKNE	Lateral femoral epicondyle
Shank	RTIB	LTIB	Mid-shank, lateral surface
Ankle	RANK	LANK	Lateral malleolus
Heel	RHEE	LHEE	Calcaneus (heel)
Toe	RTOE	LTOE	Second metatarsal head (toe)



(a) Plug-in Gait marker placement (front and back)

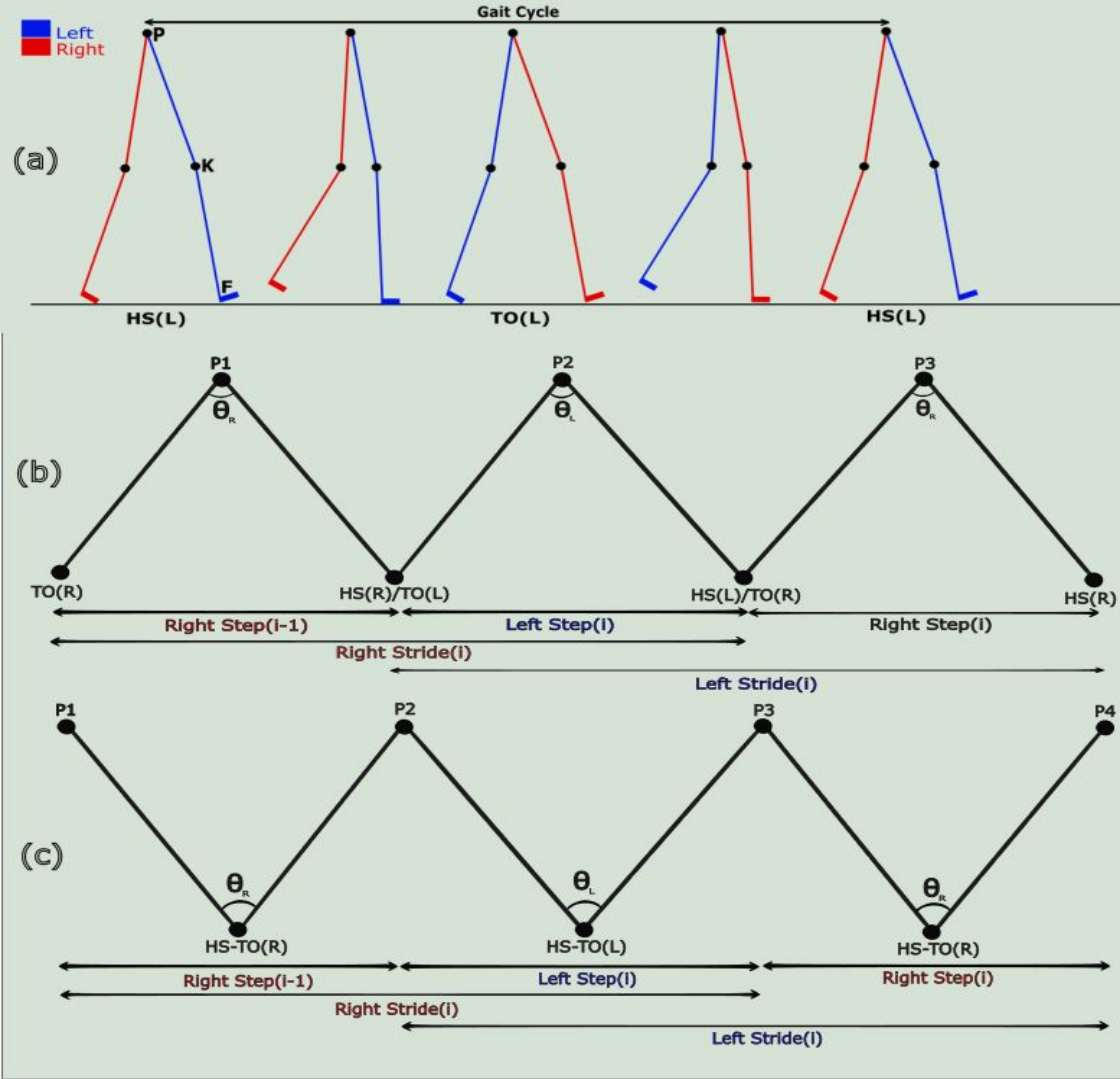


Figure 2.4: Representation of stride-length estimation models.

- (a) Full gait cycle illustrating alternating left and right limb phases.
- (b) Overground model based on **swing-phase** angular displacement
- (c) Treadmill model based on **stance-phase** inverted pendulum dynamics

Notations:

P = pelvis position;

HS = heel strike;

TO = toe-off;

θ_R , θ_L = limb excursion angles.